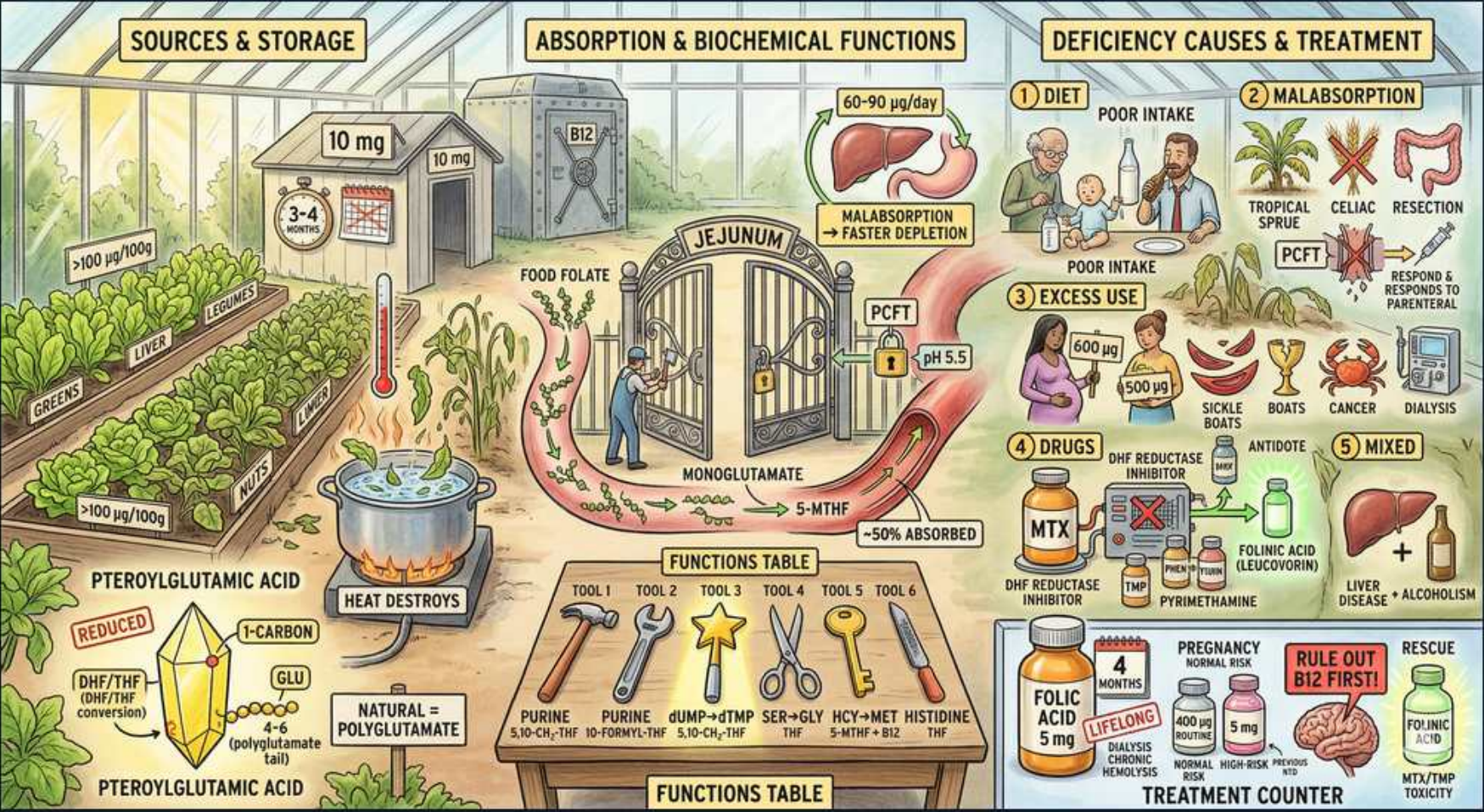


Megaloblastic — Folate Sources, Causes & Treatment



1. Green leafy sources

WHAT YOU SEE

Greenhouse of leafy greens '>100 ug/100g'

WHAT IT ENCODES

Folate (folium = leaf) comes from green leafy vegetables, legumes, liver, and nuts (>100 µg/100 g in greens); daily requirement ~400 µg (600 µg in pregnancy). In the US, grains are folate-fortified, lowering dietary deficiency. Mnemonic: folate lives in foliage.

BOARD TRAP

WRONG: assuming any well-fed adult cannot be folate deficient because of fortification. Discriminator: fortification reduces but doesn't eliminate it — alcoholics, the 'tea-and-toast' elderly, pregnant women, and malabsorbers still become deficient.

2. Heat destroys folate

WHAT YOU SEE

Pot over flame 'HEAT DESTROYS'

WHAT IT ENCODES

Folate is HEAT-LABILE — prolonged cooking/boiling destroys it, so overcooked or canned food loses much of its folate. This is a distinguishing feature from the more stable B12. Mnemonic: boil the folate away.

BOARD TRAP

WRONG: thinking diet rich in greens guarantees folate sufficiency regardless of preparation. Discriminator: heavily cooked diets can still be folate-poor; B12 is comparatively heat-stable, so cooking selectively depletes folate.

3. Small stores, fast depletion

WHAT YOU SEE

Top 'MALABSORPTION → FASTER DEPLETION', '60-90 ug/day'

WHAT IT ENCODES

Folate body stores are SMALL (~5-20 mg, lasting only 3-4 MONTHS), so deficiency develops within weeks-months of poor intake or malabsorption — far faster than B12 (years). Mnemonic: folate = small fridge (months); B12 = big vault (years).

BOARD TRAP

WRONG (swap trap): 'folate stores last years.' That's B12. Discriminator: a megaloblastic anemia appearing within months of a new insult (alcoholism, pregnancy, MTX) is far more likely FOLATE than B12 because folate stores deplete fast.

4. Absorbed in jejunum

WHAT YOU SEE

Gate 'JEJUNUM', PCFT, '~50% ABSORBED', monoglutamate

WHAT IT ENCODES

Dietary folate (polyglutamate) is deconjugated to MONOGLUTAMATE and absorbed in the proximal JEJUNUM via PCFT (proton-coupled folate transporter, optimal at acidic pH ~5.5); no intrinsic factor needed. Mnemonic: folate = jejunum + PCFT; B12 = ileum + IF.

BOARD TRAP

WRONG (favorite swap): 'folate is absorbed in the terminal ileum.' That's B12. Discriminator: jejunal disease (celiac, tropical sprue) → folate deficiency; terminal-ileal disease/resection (Crohn) → B12 deficiency.

5. Poor intake / alcoholism

WHAT YOU SEE

DIET poor intake figures, MIXED 'LIVER DISEASE + ALCOHOLISM'

WHAT IT ENCODES

The classic folate causes are poor intake states: alcoholism (poor diet + impaired hepatic folate metabolism + enterohepatic loss), the elderly, and chronic illness. Alcohol is the single most common cause of folate deficiency in developed countries. Mnemonic: alcohol attacks folate on every front — intake, absorption, storage.

BOARD TRAP

WRONG: attributing an alcoholic's macrocytic anemia to B12. Discriminator: alcohol classically causes FOLATE deficiency (and direct non-megaloblastic macrocytosis); think B12 only with neuro signs, pernicious-anemia antibodies, ileal disease, or metformin.

6. Malabsorption (celiac/sprue)

WHAT YOU SEE

MALABSORPTION 'TROPICAL SPRUE', 'CELIAC', 'RESECTION'

WHAT IT ENCODES

Folate malabsorption occurs with proximal small-bowel disease: celiac disease, tropical sprue, and jejunal resection — all hitting the jejunal PCFT absorptive site. Mnemonic: jejunal disease starves folate (ileal disease starves B12).

BOARD TRAP

WRONG: expecting terminal-ileal Crohn to primarily cause folate deficiency. Discriminator: proximal/jejunal disease (celiac, tropical sprue) → folate; distal/ileal disease (Crohn ileitis, ileal resection) → B12. Severe diffuse small-bowel disease can deplete both.

7. Increased use — pregnancy

WHAT YOU SEE

EXCESS USE 'pregnancy 600ug', sickle cell, cancer, dialysis

WHAT IT ENCODES

High-demand states deplete folate: pregnancy/lactation (need ~600 µg/day), chronic hemolysis (sickle cell — rapid RBC turnover), malignancy, and dialysis (folate is dialyzed off). Mnemonic: anything that makes cells fast (fetus, hemolysis, tumor) burns through folate.

BOARD TRAP

WRONG: not anticipating folate deficiency in a chronically hemolyzing patient (e.g., sickle cell) with a falling hemoglobin. Discriminator: high RBC turnover increases folate demand — these patients are often given prophylactic folate, and a sudden megaloblastic crisis suggests folate depletion.

8. Methotrexate (DHFR inhibitor)

WHAT YOU SEE

DRUGS 'MTX', 'DHF REDUCTASE INHIBITOR', antidote folinic acid

WHAT IT ENCODES

Antifolate drugs cause functional folate deficiency: METHOTREXATE and TRIMETHOPRIM (and pyrimethamine) inhibit dihydrofolate reductase (DHFR), blocking THF regeneration; phenytoin impairs folate absorption/metabolism. Rescue with FOLINIC ACID (leucovorin), which is already reduced and bypasses DHFR. Mnemonic: MTX/trimethoprim/pyrimethamine block DHFR → leucovorin rescues.

BOARD TRAP

WRONG: trying to reverse methotrexate toxicity with plain FOLIC acid. Discriminator: folic acid still needs DHFR (which MTX has blocked) to become active — only FOLINIC acid (leucovorin), already reduced, bypasses the block and rescues the patient.

9. Pregnancy supplement / NTD

WHAT YOU SEE

TREATMENT 'FOLIC ACID 4 weeks', 'PREVENT NTD'

WHAT IT ENCODES

Treat folate deficiency with oral folic acid (~1-5 mg/day; the megaloblastic anemia corrects over weeks). PERICONCEPTIONAL folate (≥400 µg/day, 4 mg in high-risk) prevents neural-tube defects (spina bifida, anencephaly) — start before conception since the tube closes by ~week 4. Mnemonic: folate before you're pregnant prevents NTDs.

BOARD TRAP

WRONG: starting folate only after a positive pregnancy test to prevent NTDs. Discriminator: the neural tube closes very early (~28 days), often before pregnancy is recognized — folate must be PERICONCEPTIONAL/started before conception to be protective.

10. Rule out B12 first

WHAT YOU SEE

Brain icon 'RULE OUT B12 FIRST!'

WHAT IT ENCODES

Before giving folate for any macrocytic/megaloblastic anemia, EXCLUDE B12 deficiency. Folate alone corrects the anemia but does not treat — and can worsen/unmask — B12 neurologic disease (subacute combined degeneration), which then progresses silently. Mnemonic: folate fixes the blood but hides the danger to the nerves.

BOARD TRAP

WRONG (high-yield trap): giving folate empirically for a macrocytic anemia. Discriminator: always check B12 first; folate monotherapy in occult B12 deficiency masks the CBC while subacute combined degeneration advances. Replace B12 (or co-treat) when both are uncertain.

11. DHF/THF & 1-carbon

WHAT YOU SEE

Crystal 'DHF/THF', '1-CARBON', polyglutamate

WHAT IT ENCODES

Folate's job is to carry ONE-CARBON units for nucleotide synthesis (thymidylate/purines) and methylation. THF is the active workhorse; tissue folate is stored as POLYglutamate (trapped inside cells), absorbed as monoglutamate. Impaired DNA synthesis is why deficiency causes megaloblastic change. Mnemonic: folate = the one-carbon delivery van for DNA.

BOARD TRAP

WRONG: assuming folate and B12 act on the same biochemical step. Discriminator: folate provides the 1-carbon units for DNA; B12 additionally runs methylmalonyl-CoA mutase (myelin) — which is why only B12 deficiency causes neurologic disease.

12. Folinic acid rescue

WHAT YOU SEE

'FOLINIC ACID (LEUCOVORIN)'

WHAT IT ENCODES

Folinic acid (leucovorin) is a reduced folate that enters the active pool WITHOUT needing DHFR, so it rescues cells from DHFR-inhibitor toxicity (methotrexate, trimethoprim, pyrimethamine). 'Leucovorin rescue' is timed after high-dose MTX to spare normal cells. Mnemonic: leucovorin = the antidote that skips the blocked enzyme.

BOARD TRAP

WRONG: substituting folic acid for folinic acid in MTX rescue, or giving leucovorin too early (which would also rescue the tumor). Discriminator: only FOLINIC acid bypasses the DHFR block; in chemotherapy it is timed to protect normal tissue while still letting MTX act on the cancer.